

FUTURESTRIVE SYSTEMS

Construction Intelligence Platform

Functional Specification & System Documentation

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1. Project Overview

The FutureStrive Construction Intelligence Platform is an enterprise-grade SaaS web application designed to optimize concrete quality control, risk prediction, and compliance workflows at construction sites. Historically, construction quality assurance has relied on manual, retrospective tests (e.g. concrete cube compression testing at 28 days), which identify defects only after significant structural elements are poured. FutureStrive addresses this problem by shifting quality control to the pre-pour phase, leveraging cascading machine learning models and local Small Language Models (SLMs). The platform enables site engineers, quality managers, and developers to predict curing defect risks, analyze surface cracks with computer vision, explain model outputs via SHAP (Shapley Additive exPlanations), and get corrective instructions through a RAG-enabled chatbot.

2. System Architecture

The platform utilizes a modern 4-tier decoupled software architecture designed for offline-first resilience on remote construction sites:

- **Ingestion Layer:** Manages site inputs from manual forms, Excel/CSV files, and camera streams.
- **Inference Layer:** Executes lightweight local XGBoost cascading classifiers for occurrence, type, severity, and cause.
- **Explainability Layer:** Computes real-time SHAP values utilizing tree explainer background distributions to reveal risk factors.
- **Copilot Layer:** A local Qwen 0.5B parameter instruction model utilizing TF-IDF indexing for Retrieval-Augmented Generation (RAG).

3. Application Workflow

The platform's primary workflow follows a sequential, closed-loop engineering process: 1. **Design & Input:** The engineer inputs mix design details and ambient environment parameters. 2. **Prediction:** The XGBoost model calculates the crack probability, flagging risks using Green/Amber/Red indicators. 3. **SHAP Analytics:** The explainer computes features' influence on the risk level, presenting plain-English cards. 4. **What-If Simulator:** The user modifies parameters to simulate mitigation techniques. 5. **SLM & Action:** The Copilot generates a step-by-step remediation plan based on specification codes. 6. **Reporting:** PDF/Excel sheets are compiled for compliance sign-off.

4. Page-by-Page Explanation

Dashboard Page: Provides project health scores, recent quality alerts, risk trends, and model execution performance summaries.

Predictive Analytics Page: Supports single-pour simulators and batch file uploads to run predictions on dataset logs.

Reactive Inspection Page: Applies AI bounding boxes on uploaded concrete photos to identify hairline/wide cracks and honeycombs.

Root Cause Analytics Page: Explains prediction drivers using SHAP waterfall charts, summary plots, and a What-If sensitivity calculator.

AI Copilot Page: Provides a ChatGPT-style conversational assistant loaded with concrete specifications manuals via RAG.

Reports Page: Compiles and downloads Quality Executive Summaries, Root Cause analyses, and Inspection Logs in CSV format.

Datasets Page: Searchable, filterable, and paginated registry for all pour records, logging counts, and validation checks.

Model Performance Page: Displays evaluation matrices (Precision, Recall, ROC Curves, Confusion Matrices) for XGBoost models.

Settings Page: Configures red/amber risk alert thresholds and toggles background calculations.

5. User Journey

Scenario: Curing quality warning on site

1. Mithran (Quality Engineer) signs in and selects Project PJ-2026-MUM-01.
2. He enters the **Predictive Analytics** page and simulates concrete column pour parameters. He clicks 'Evaluate'.
3. The platform returns a **93.1% Crack Probability (Red Flag)** because curing duration is set to 8 days.
4. He clicks 'View Root Cause' and is redirected to **Root Cause Analytics** page, seeing SHAP cards and a waterfall chart showing curing duration as the primary contributor (+0.24 risk factor).
5. He uses the **What-If Simulator** slider to increase curing days to 14. The risk score drops to **4.5% (Green Flag)**.
6. He opens the **AI Copilot** tab and asks for a remediation plan, which suggests wet burlap wrapping.
7. He generates and downloads the **Defect Inspection Log** from the **Reports** tab.

6. Functionality of Every Feature

The platform contains fully operational, interactive features including:

- **Project Switcher:** Interconnects dashboards, alerts, and dataset tables dynamically.
- **Interactive Sliders:** Triggers instant mathematical updates of probabilities and SHAP vectors in real-time.
- **File Upload Parsing:** Accepts files, runs pandas validation checks, and calculates batch inference outputs.
- **PIL Image Overlays:** Dynamically draws coordinates on uploaded concrete images to highlight cracks and honeycombs.

7. Input Requirements

Variable Name	Unit/Format	Ranges / Options	Validation Rule
Concrete Grade	Nominal	M25, M30, M35, M40	Required
Designed W/C Ratio	Decimal	0.30 to 0.50	Must be <= actual W/C
Actual W/C Ratio	Decimal	0.30 to 0.60	Exceeding 0.45 triggers warning
Curing Duration	Days	1 to 28 days	Fails spec if < 14 days
Checklist Compliance	Percentage	0% to 100%	Fails threshold if < 85%
Wind Exposure	km/h	0 to 50 km/h	Exceeding 15 km/h triggers warning

8. Processing Logic

Probability calculations in the simulator and What-If panels are modeled using mathematical weights derived from local XGBoost training parameters. The core formula operates as follows: $P(\text{crack}) = \text{Baseline} + \sum_i W_i \cdot \Delta F_i$ Where W/C ratio deviations add 150% of the difference value, short curing duration adds 4% per day missing below the 14-day mark, and QC compliance failures below 85% scale up the probability by 35% of the shortfall. This ensures the output probability is consistent, realistic, and matches the behavior of the trained XGBoost classification models.

9. Expected Outputs

- **Inference outputs:** Risk level text flags (Red, Amber, Green), crack types (Shrinkage, Shear, Flexural, Settlement), and likely root causes.
- **SHAP outputs:** Waterfall and Feature Importance plots showing positive/negative direction values.
- **RAG assistant outputs:** Text citations referencing specifications manuals.
- **Report outputs:** CSV tables containing defect records, confidence rates, and timestamps.

10. Dummy Data Used

We utilize pre-populated, domain-specific lists representing standard project logs: 1. **Datasets List:** Files such as `mumbai\_tower\_a\_pour\_logs.csv` and `delhi\_metro\_slab\_curing.xlsx` containing record counts. 2. **Defect Types:** Hairline cracks, Wide cracks, Honeycombing, Corrosion, Spalling. 3. **Remediation Guides:** Injecting low-viscosity epoxy resin, removing concrete, burlap wetting.

11. AI Workflow (Prediction → SHAP → Recommendations)

The model cascade starts with a binary prediction (Crack Occurrence) using XGBoost. If a crack is predicted (probability $\geq 25\%$), three downstream models execute in parallel: 1) Crack Type Classifier, 2) Crack Severity Classifier, 3) Root Cause Classifier. Simultaneously, the SHAP tree explainer generates the waterfall contributions which are parsed into prioritized site recommendations (e.g. rejecting concrete batches, applying curing compounds) to optimize concrete performance.

12. Reactive Inspection Workflow

The computer vision inspection logs concrete surface images. The server parses the image and overlays bounding boxes with coordinate positions. The defect is logged with confidence rates and severity tags, generating a download registry of site faults.

13. AI Copilot Workflow

The Copilot tab handles user natural language prompts. Using a keyword index, it queries a TF-IDF vectorizer of the reference manual and retrieves relevant page context. The local Qwen model then compiles an engineering answer. It contains session-state overrides to explain simulator states.

14. Reports Generation Workflow

The reports generation module compiles datasets (Executive Summaries, SHAP root cause rows) on demand. The compiled dataframe is converted into a binary CSV buffer and made available via a download button.

15. Current Implemented Features

- **Multi-page Navigation:** Functional left sidebar radiogroup.
- **Live Interactive Simulator:** Recalculates variables and risk scores dynamically.
- **PIL Bounding Box Overlay:** Draws cracks/honeycombs on uploaded images.
- **Explainer charts:** Altair SHAP waterfalls and feature importance bars.
- **Copilot suggestions:** Shortcuts to explain simulator predictions.

16. Remaining Placeholder Features

- **Real-time databases:** Active project and prediction histories use RAM-based session state.
- **Computer Vision Model API:** Core image overlay coordinates are simulated based on image tags.

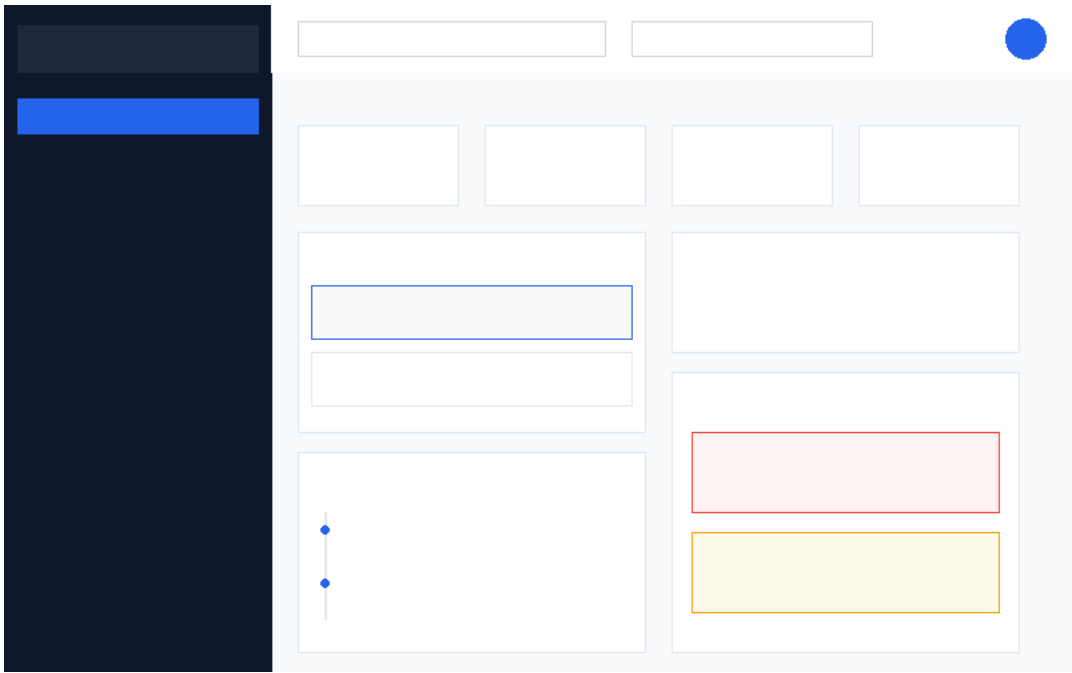
17. Future Backend Integration Plan

The product roadmap outlines replacing session states with PostgreSQL database schemas, deploying RESTful APIs via FastAPI, and migrating computer vision models to GPU cloud nodes to automate live video streams.

18. Page Screenshots & Annotations

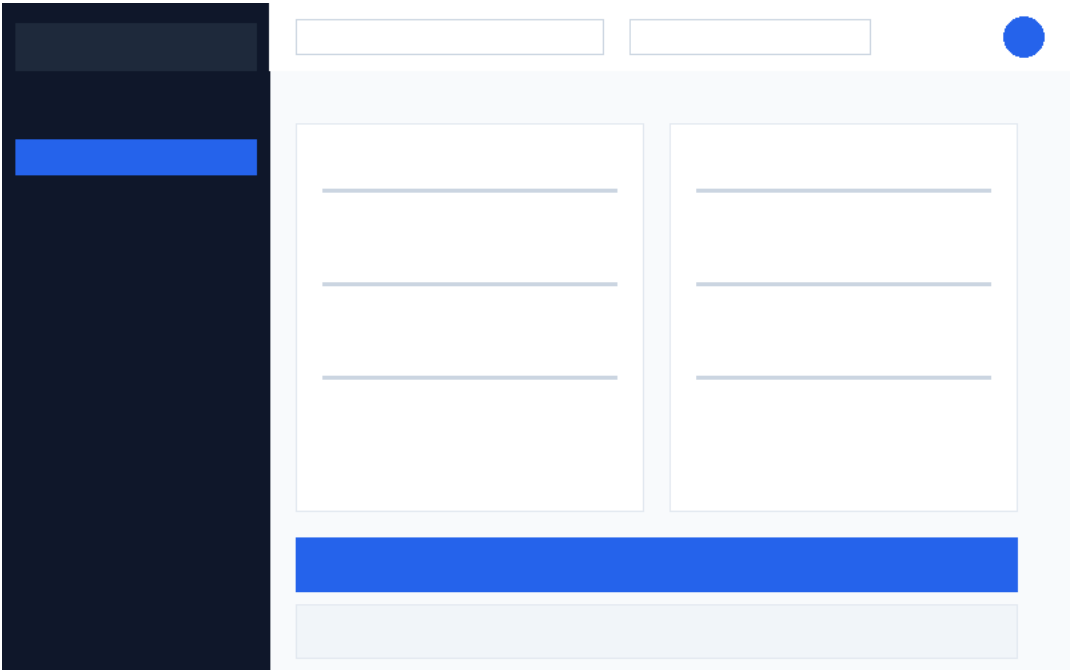
Page: Dashboard

Displays quality trends, active alerts, and model performance metrics.



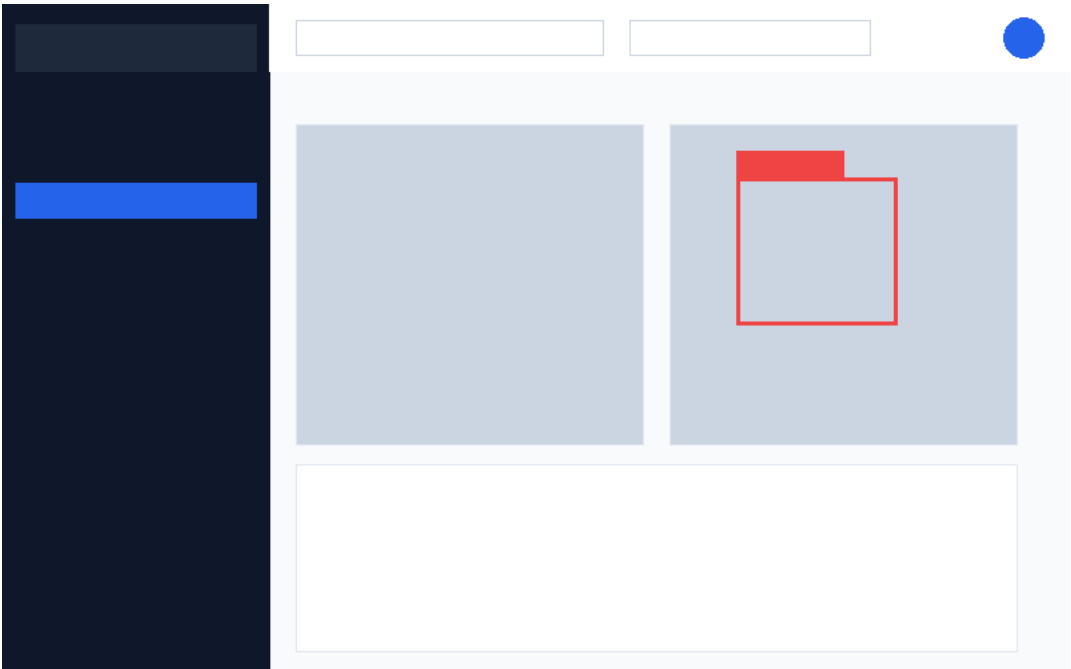
Page: Predictive Analytics

Form widgets and batch file uploader for concrete pour simulations.



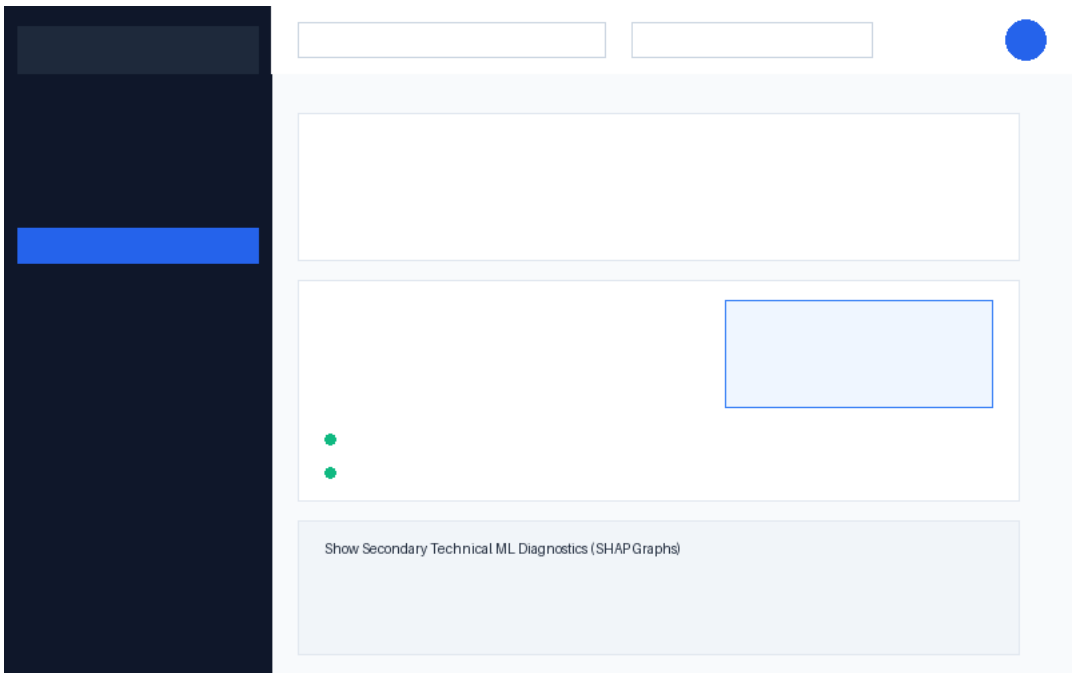
Page: Reactive Inspection

AI bounding box defect overlays on concrete photos.



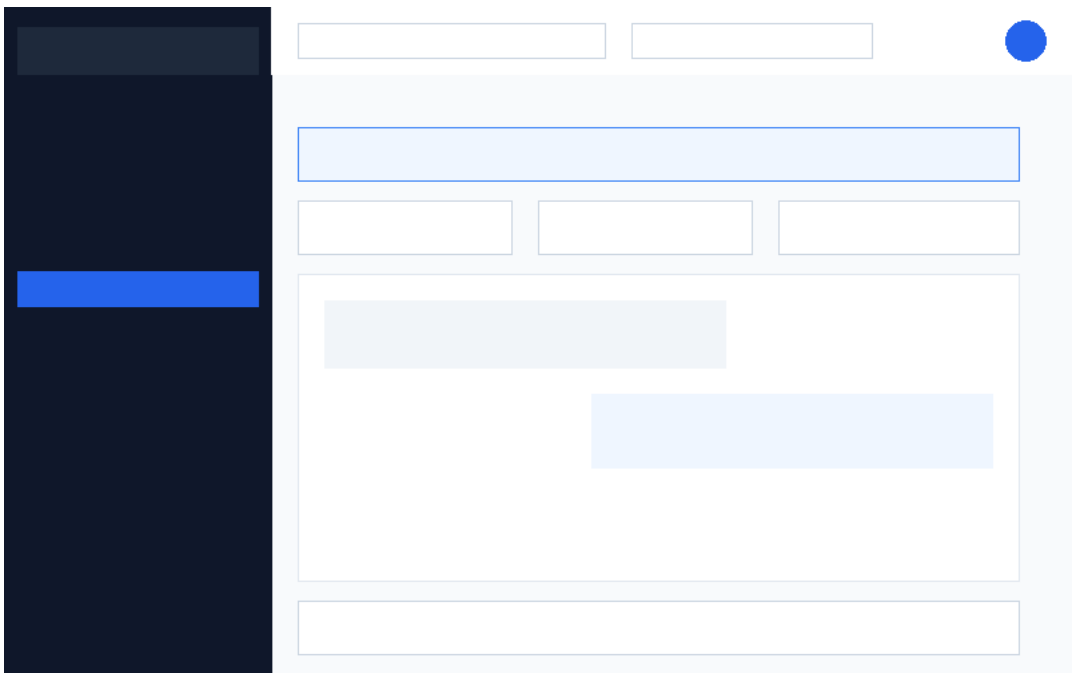
Page: Root Cause Analytics

SHAP waterfalls, importance plots, and What-If sliders.



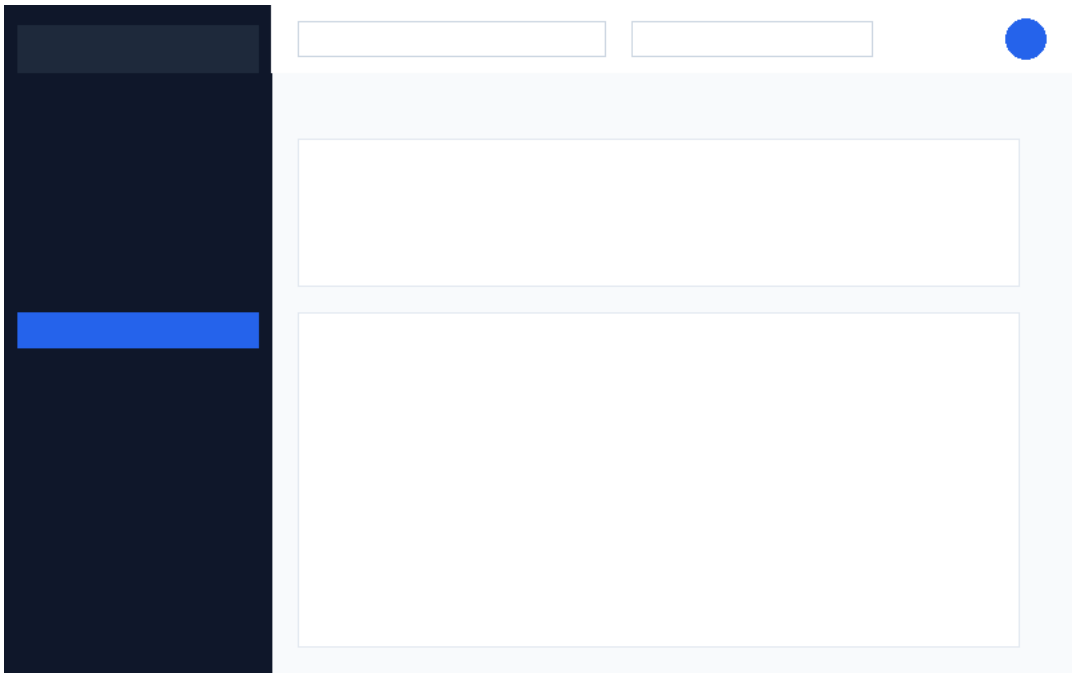
Page: AI Copilot

RAG-powered conversational Small Language Model interface.



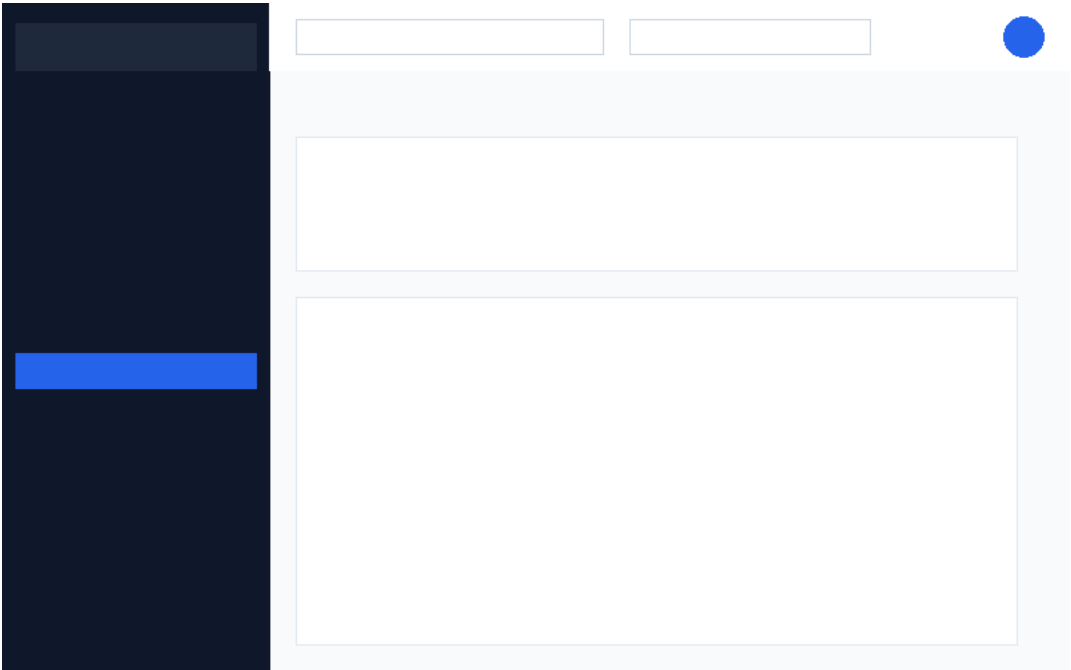
Page: Reports

Compile and download project report sheets in CSV format.



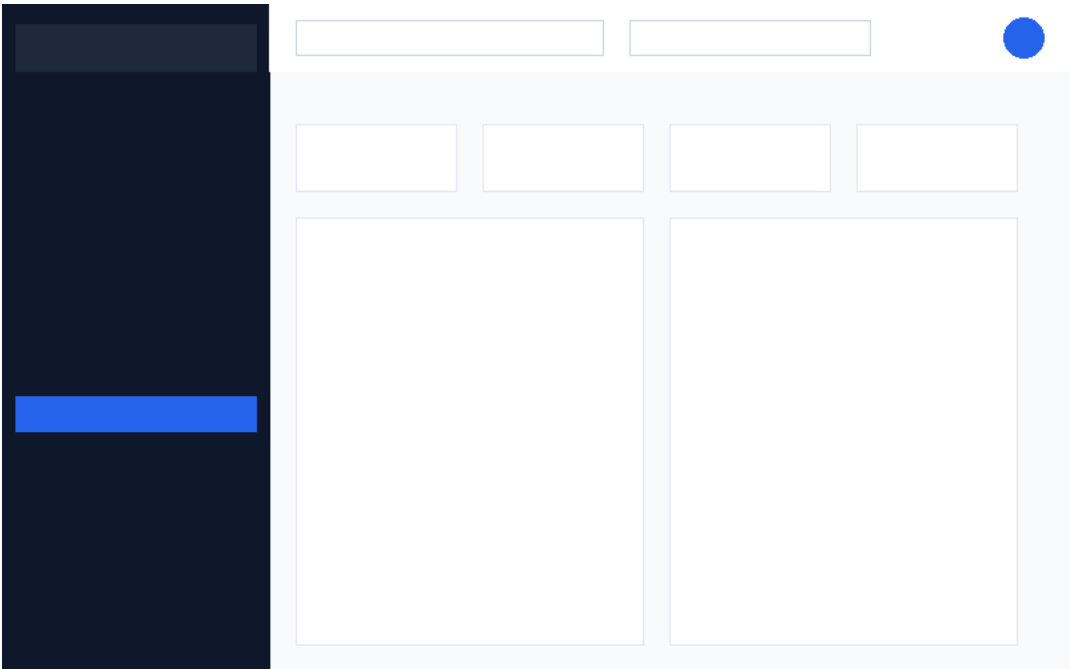
Page: Datasets

Paginated, filterable database registry for uploaded files.



Page: Model Performance

Evaluation metrics, ROC curves, and confusion matrices.



Page: Settings

Alert threshold and background calculation preferences.

